# S4DR-Net: Self-Supervised Spatial–Spectral Distance Reconstruction Network for Multispectral Point Cloud Classification

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***Abstract*—Multispectral LiDAR point clouds are valuable in remote sensing for their spatial–spectral consistency, yet their high acquisition and annotation costs pose significant challenges. To mitigate this, self-supervised learning has emerged as a promising solution, reducing reliance on annotated data while improving model generalization. However, existing self-supervised frameworks for point clouds often overlook the complexity of ground object distribution in large-scale remote sensing scenarios and fail to leverage the spectral information inherent in multispectral point clouds. In this article, we introduce the self-supervised spatial–spectral distance reconstruction network (S4DR-Net), a novel self-supervised pretraining network designed for multispectral point cloud classification. Serving as the key component of the network, the spatial–spectral distance prediction module (S-SDP) effectively addresses these limitations by reconstructing the distance relationships between voxel blocks in 3-D Euclidean as well as spectral spaces. By jointly considering spatial and spectral distances, S-SDP enables the network to learn a unified representation that captures the intrinsic spatial–spectral consistency of multispectral point clouds. This design allows S4DR-Net to generate low-dimensional feature representations in a self-supervised manner, without reliance on manual annotations. We conducted experiments and evaluated on two real-world multispectral point cloud datasets. The results demonstrate that S4DR-Net consistently outperforms existing self-supervised pretraining methods, achieving superior accuracy and generalization compared with current state-of-the-art approaches. The code will be released at https://github.com/KustTeamWQW/S4DR-Net**

***Index Terms*—Model pretraining, multispectral LiDAR, point cloud classification, self-supervised learning.**

## I. Introduction

POINT clouds contain rich geometric structural information, while multispectral and hyperspectral images provide abundant spectral data [1], [2], [3]. In the field of remote sensing, several studies have combined the strengths of both to enhance 3-D environmental perception [4]. However, due to inconsistencies between images and point clouds, it has been challenging to fully leverage spatial and spectral information. Furthermore, multispectral point clouds provide consistent and precise 3-D spatial data, as well as rich spectral information. This makes them highly valuable for large-scale remote sensing scene interpretation. This holds significant potential for applications across various fields, including forest monitoring [5], coastal management [6], and land cover mapping [7].

In recent years, deep learning techniques applied to multispectral LiDAR data, i.e., multispectral point clouds, have made significant strides. Notable advances include point-based methods [8], voxel-based methods [9], and graph-based methods [10]. Although these methods have proven effective in the field of multispectral point cloud classification, most of them rely heavily on large amounts of labeled data [11], [12]. To address this challenge, scholars have recently proposed self-supervised pretraining methods for point clouds to improve model generalization and reduce model dependence on labeled data. These types of methods mainly learn general features from large amounts of point cloud data through proxy tasks, which can be broadly categorized into two categories: contrast-based methods [13], and reconstruction-based methods [14], [15]. Contrast-based methods, such as Point-GCC, learn by contrasting similar and dissimilar samples [13]. These methods bring similar samples closer in feature space while pushing dissimilar ones apart, effectively capturing global structures. Reconstruction-based methods focus on learning to reconstruct input data by predicting missing points or structures. For example, Voxel-masked autoencoder (MAE) masks a portion of the input point cloud and reconstructs the missing data, prompting the model to learn meaningful geometric features from the unmasked points [14]. This method effectively captures local features.

Although these methods have yielded impressive results in indoor point cloud classification, most self-supervised pretraining models still face challenges when transferring to multispectral point clouds in remote sensing scenarios. We think this issue arises from two key factors. First and foremost, traditional frameworks fail to fully exploit spectral information and overlook the inherent spatial–spectral consistency of multispectral point clouds, hindering their ability to capture the relationships between spatial and spectral features.

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This limitation represents the primary challenge when applying these methods to multispectral point clouds. Moreover, unlike indoor point clouds, remote sensing point clouds exhibit greater complexity and diversity. They span a wide range of spatial scales, from small objects to large areas, and have highly irregular point density distributions, with sparse regions such as open fields and dense areas like urban environments. In addition, the distribution of ground objects in remote sensing scenarios is far more intricate, with various land cover types, further contributing to the increased variability. Due to these pronounced differences in spatial and contextual characteristics, directly transferring methods from indoor environments to remote sensing scenarios remains highly challenging.

To address the aforementioned issues, this article proposes a self-supervised spatial–spectral distance reconstruction network (S4DR-Net) for multispectral point cloud classification in remote sensing scenarios. The framework consists of two main components: a voxel spatial feature residual module, which enhances the extraction of local geometric features, and a set of autoencoders, which perform distance reconstruction under self-supervised guidance. We first voxelize the multispectral point clouds and then perform graph modeling. By leveraging the inherent spatial–spectral consistency of multispectral point clouds, we introduce a custom rule to calculate the spatial–spectral distance between voxel blocks after graph modeling. Furthermore, we design the spatial–spectral distance prediction module (S-SDP), embedded within the autoencoders, which employs this distance as a supervisory signal to guide the reconstruction of voxel relationships in both 3-D Euclidean space and spectral space. Our main contributions are as follows.

1) We propose the first self-supervised pretraining method, S4DR-Net, for multispectral point cloud classification in remote sensing scenarios. The S-SDP module acts as a plug-and-play component, guiding the pretraining process for multispectral point cloud classification.
2) We introduce a novel similarity metric for multispectral point cloud voxel blocks, based on the intersection of spectral channel color histograms. This innovative metric enables effective self-supervised signal modeling from intrinsic spatial–spectral information, addressing the key limitation of traditional point cloud self-supervised methods in fully exploiting the spatial–spectral consistency of multispectral point clouds.

## II. Related Work

### A. Multispectral Point Cloud Classification

In recent years, with the advancement of deep learning, multispectral point cloud classification methods have evolved into three main categories: voxel-based, point-based, and graph-based methods. For voxel-based methods, point cloud data is inherently unordered and irregular, making it unsuitable for direct processing by traditional CNN frameworks designed for image data [16]. To adapt to CNN architectures, multispectral LiDAR data rasterized into 2-D images, enabling feature extraction via standard 2-D convolutional layers for land-cover classification [7], [17]. However, this method results in a significant loss of spatial information inherent to the original point cloud structure. To address this limitation and be compatible with traditional CNN architectures, the 3-D U-Net employed 3-D convolutional neural networks (3-D CNNs) to extract features from voxelized point clouds [18]. While this approach preserves some spatial context, it often leads to a loss of fine details and incurs high computational costs [19]. By regularizing point clouds into voxel grids, voxel-based methods rely on 3-D CNNs for feature extraction, but their primary challenge remains overcoming computational bottlenecks associated with dense convolutions. Current research focuses on optimizing sparse convolution techniques and feature aggregation strategies to achieve both efficient and accurate 3-D perception.

To mitigate the loss of information during voxelization and reduce the computational overhead of 3-D CNNs, point-based methods have emerged as an alternative for point cloud processing. PointNet++ introduced an MLP-based approach for feature extraction, leveraging the geometric invariance of point clouds to aggregate multiscale features [20]. Building on this, SE-PointNet++ incorporated a squeeze-and-excitation (SE) block, enabling the model to adaptively adjust to varying channel features [21]. This joint modeling of spatial and spectral information has proven effective for multispectral point cloud classification. Following the success of MLP-based methods, PointMLP proposed a deep hierarchical residual MLP architecture that abandons traditional reliance on local geometric feature aggregation. This design results in a lightweight yet efficient framework for point cloud classification [22]. Inspired by the transformer architecture's long-range information modeling capabilities, PointTransformer applied it to point cloud processing to capture dependencies across distant points in the cloud [23]. However, despite its parallel computation potential, the quadratic complexity of the transformer leads to higher computational costs [24]. To address this, researchers have integrated the Mamba architecture into point cloud processing [25]. The Mamba architecture can perform linear complexity modeling, improving computational efficiency and reducing overhead. Recently, 3-D-UMamba combined Mamba with the U-Net network, resulting in a generalized and lightweight point cloud classification framework [26]. VMamba was established as an efficient vision backbone by combining state-space models with 2-D selective scanning, resulting in linear-time visual representation learning for remote sensing data [27]. Spectral informed Mamba (SI-Mamba) integrated Mamba state-space models with MAEs and spectral graph theory, resulting in a robust framework for point cloud processing that significantly improves performance in classification, segmentation, and few-shot learning tasks [28]. Point-based methods have thus evolved from local operators like PointNet++ to more global perceptual architectures such as transformer and the Mamba state space model, facilitating significant advancements in fine-grained 3-D recognition tasks through adaptive relationship mining.

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Point cloud data is inherently non-Euclidean. This makes it naturally amenable to graph-based approaches. Such approaches can effectively capture topological structures and are well-suited for handling unordered data [29], [30], [31]. MKGSL introduced multikernel graph structures, allowing the network to adaptively capture high-dimensional and implicit topological relationships [32]. In subsequent work, MGC2N incorporated a self-attention masking mechanism during graph modeling, allowing the network to dynamically adjust the extracted topological information and provide each point with its corresponding contextual data [33]. In addition, MGC2N introduced a cross-convolution module that simultaneously extracts global spatial distribution features and local spectral features of ground objects, significantly improving multispectral point cloud classification accuracy. By inherently integrating spectral attributes and spatial topology, graph-based methods have become a dominant approach in multispectral point cloud processing.

### B. Self-Supervised Pretraining of Point Cloud

In recent years, following the success of self-supervised pretraining in natural language processing and image processing, researchers have turned their attention to applying similar methods to point clouds. Self-supervised pretraining for point clouds is a label-free deep learning approach that allows models to autonomously uncover intrinsic relationships and latent information within the data through specially designed proxy tasks, all without relying on annotated data. This enables the extraction of general features specific to a given point cloud scenario at a low cost through pretraining.

Point cloud self-supervised learning methods are classified into reconstruction methods and contrast methods based on the proxy tasks they perform. Reconstruction methods, such as Point-MAE [34], used PointNet as the feature extraction backbone and adopted a random high-proportion masking strategy in each sample to select the point cloud to be reconstructed, utilizing the spatial position constraints of the masked point cloud for self-supervised learning. Inspired by the global context modeling capabilities of transformer, Point-BERT introduced it as a feature extraction backbone to reconstruct point clouds after random masking [35]. OcCo generated occluded point clouds from randomly masked images [36]. Methods based on contrast, such as PointContrast [37], obtained general features by learning the potential similarities between different point cloud frames in the same scene through contrast [38]. In addition, some researchers extracted general features by guiding models to predict the inherent composition structure in specific scenarios. For example, DHGCN explored the hidden topological relationships between data by reconstructing the spatial distance relationships between point cloud voxel blocks [15]. TG-ADet, a terrain-guided network for airborne laser scanning (ALS) point clouds, offers valuable insights that can be adapted to multispectral contexts. TG-ADet introduces a multistage terrain-guidance (MTG) module that predicts terrain distribution and guides detection stages, effectively suppressing background interference in complex environments [39].

Although the methods mentioned above have made significant progress in self-supervised point cloud representation learning, they fail to fully leverage the spatial–spectral consistency inherent in multispectral point clouds. For instance, while Point-MAE learns geometric features through mask coordinate prediction, it does not incorporate spectral channels into the reconstruction target. Moreover, existing pretraining methods do not account for the diversity and the complexity of geospatial object distributions in remote sensing scenarios. As a result, these methods struggle to be directly applied to multispectral point clouds in remote sensing scenarios. To overcome these challenges, this article introduces the first self-supervised pretraining method specifically tailored for multispectral point cloud classification in remote sensing scenarios.

## III. Methodology

In this article, we propose a new self-supervised pretraining method, S4DR-Net, for multispectral point cloud classification in remote sensing scenarios. The overall architecture is illustrated in Fig. 1. The self-supervised pretraining stage adopts a dual-path structure consisting of a voxel spatial feature residual module and autoencoders, which perform distance reconstruction using self-supervised signals. The autoencoders consist of three voxel feature extraction layers, each containing an S-SDP module for distance reconstruction. The specific structure of the S-SDP module is shown in Fig. 2.

### A. Overall Architecture

Multispectral point clouds are represented as $X = [x, y, z, \lambda_1, \ldots, \lambda_L] \in \mathbb{R}^{(L+3)\times N}$, where the spatial information is denoted as $f_{\text{spatial}} = [x, y, z]$, and the spectral information is denoted as $f_{\text{spectral}} = [\lambda_1, \ldots, \lambda_L]$. $L$ is the number of spectral bands and $N$ is the number of points. First, due to the inherent spatial–spectral consistency of multispectral point clouds, we design the network as a dual-branch structure. We obtain spatial features through feature slicing, denoted as $f_{\text{spatial}}$. Subsequently, feature extraction is performed separately on $f_{\text{spatial}}$ and connected to the subsequent feature extraction layer through a residual connection to prevent the network from ignoring the learning of the original spatial features of the point clouds. The specific processing method is as follows:

$$\overline{\mathbf{F}}_V = \frac{1}{N_V}\sum_{p_n \in V} \mathbf{F}_n, \quad \mathbf{F}_n = \left[\lambda_1, \lambda_2, \ldots, \lambda_C\right]^{\top} \in \mathbb{R}^C \tag{1}$$

$$\tilde{f}_{V_\text{spatial}} = \text{MLP}\left(\overline{\mathbf{F}}_V\left(f_{\text{spatial}}, \mathcal{V}(p_i)\right)\right) \tag{2}$$

where $p_n$ is the points contained in each voxel, $N_V$ is the number of point clouds corresponding to each voxel, $\mathbf{F}_n$ is the number of point clouds corresponding to each voxel, and $\overline{\mathbf{F}}_V$ indicates the operation of averaging all points within a voxel using an index. For another branch, which consists of three feature extraction layers, each layer returns its prediction of the shortest path in space based on the extracted features to achieve self-supervised learning. At the beginning of each layer, the point cloud data passes through the feature enhancement (FE) module. This process begins with the KNN

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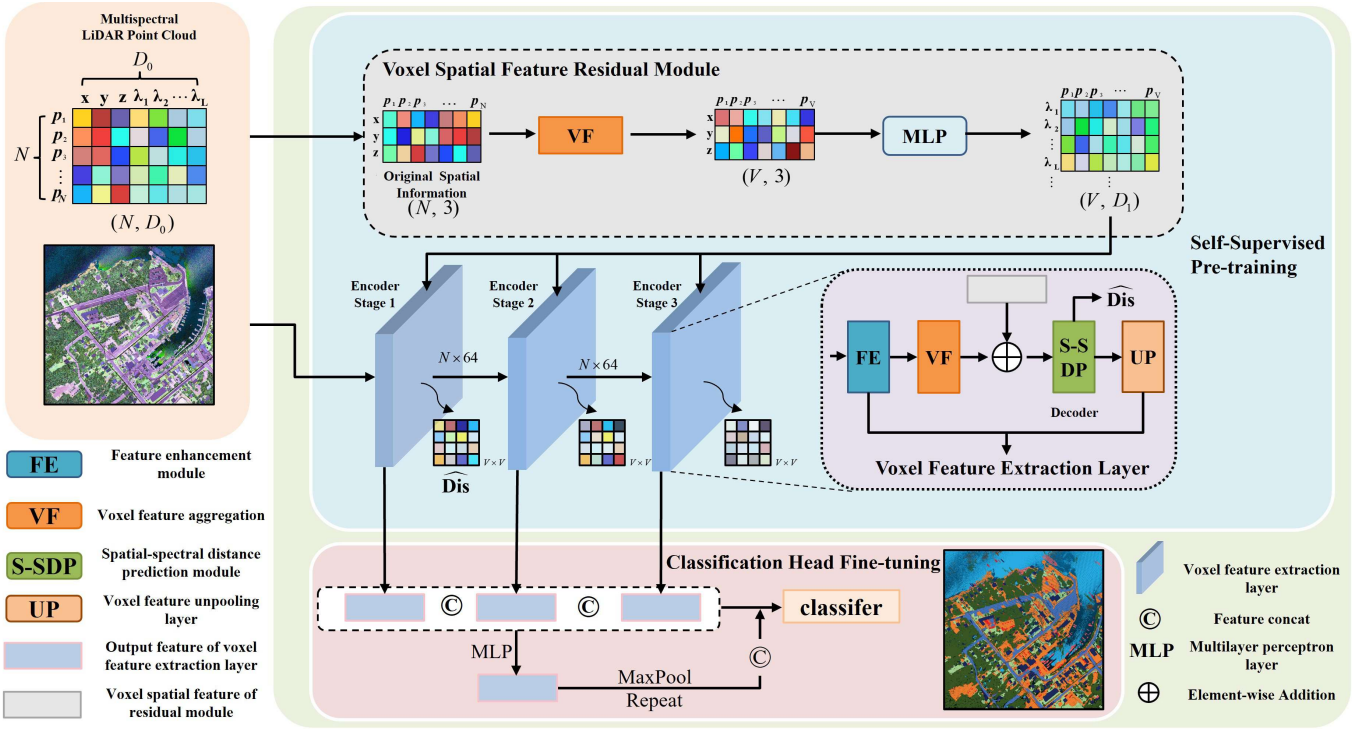



Fig. 1. Framework of the proposed S4DR-Net, which consists of self-supervised pretraining and classification head fine-tuning. The self-supervised pretraining is a dual-branch network, including three voxel feature extraction layers and a voxel spatial feature residual module with residual connections. The S-SDP module in each voxel feature extraction layer performs distance reconstruction based on the extracted features. The classification head fine-tuning is responsible for loading the pretrained model and fine-tuning it for multispectral point cloud classification.

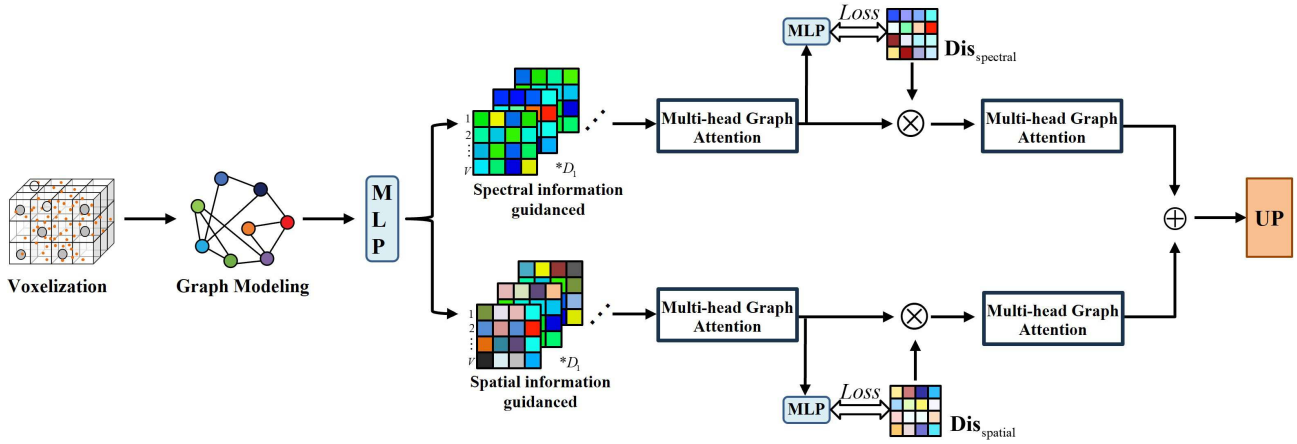



Fig. 2. S-SDP module. The module first performs graph modeling on voxelized point cloud blocks. After being mapped into high-dimensional space through an MLP, the data is fed into a dual-branch structure, each branch performs distance reconstruction based on the spatial and spectral information, guided by the self-supervised signals $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$ derived from our preprocessing. The output features are the sum of the features from both branches, which serve as the input for the subsequent unpooling layer.

algorithm querying to gather the neighboring points for each point. The features of the central point are then concatenated with the feature distances between it and each of its neighbors. This combined feature set is subsequently mapped to a higher dimensional space via an MLP. Finally, the extracted local information is aggregated to the central point through a max-pooling operation. The specific details of this process are elaborated in Section III-D. Based on the mapping relationship $\mathcal{V}(p_i)$, point features are mapped to voxel features for processing. Then, the shortest path length is predicted through the S-SDP module. Finally, the features undergo unpooling to convert them back to point-level representations, which serve as input to subsequent network layers. The overall algorithm process is as follows:

$$\tilde{f}_i = \mathrm{FE}\left(\mathrm{KNN}(f_i), f_i\right) \tag{3}$$

$$\tilde{f}_V = \overline{\mathbf{F}}_V\left(\tilde{f}_i, \mathcal{V}(p_i)\right) \tag{4}$$

$$\mathrm{Unpool}\left(f_v, \mathcal{V}(p_i)\right) = \sum_{k=1}^{V} (f_v)_k \cdot \mathcal{V}(p_i)_k \tag{5}$$

$$\text{output} = \mathrm{Unpool}\Big(\text{S-SDP}\big(\tilde{f}_V \oplus \tilde{f}_{V_\text{spatial}}\big)\Big) \oplus \tilde{f}_i \tag{6}$$

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where $f_v \in \mathbb{R}^{C\times V}$ represents the voxel features before unpooling, $\mathcal{V}(p_i) \in \mathbb{R}^{N\times V}$ represents the Boolean indexing from point cloud points to voxel blocks, and $V$ represents the number of voxel blocks (125 in this article).

### B. Voxelization and Graph Modeling

In this article, we use the shortest path length between voxel blocks in both Euclidean and spectral spaces as a proxy task, as we believe that the length of the shortest path between blocks cannot only serve as a self-supervised signal to guide the model in learning effective neighborhood information between multispectral point cloud voxel blocks while excluding redundant information, but also guide the model in learning multiscale long-range neighborhood information between voxel blocks. These processes stimulate the model to self-learn the implicit topological structure between multispectral point cloud voxel block data, thereby achieving the effect of pretraining.

Specifically, we process the raw point cloud data in blocks. For each block of multispectral point cloud data, we perform centering, normalization, and sampling operations. Since multispectral point cloud data has consistent spatial–spectral information, we extract the spatial feature channel and spectral feature channel of the point cloud data for separate processing.

For the spatial information of point clouds, we first need to divide the multispectral point cloud data into point cloud voxel blocks. We divide the resulting block data proportionally across three spatial dimensions (the split number used in the experiment is five). We then remove redundant voxel blocks, such as those that do not contain any point cloud points, to obtain the final voxel blocks, saving the index between points and voxels, as shown in the formula

$$\mathcal{V}(p_i) = \left( \left\lfloor \frac{x_i - x_{\min}}{v_x} \right\rfloor, \left\lfloor \frac{y_i - y_{\min}}{v_y} \right\rfloor, \left\lfloor \frac{z_i - z_{\min}}{v_z} \right\rfloor \right) \tag{7}$$

where $v_x$, $v_y$, and $v_z$ represent the split number. $\mathcal{V}(p_i)$ represents the mapping relationship between point clouds and voxel blocks. We defined a rule: if two voxel blocks overlap after being enlarged (1.2 times in this article), they are considered spatially adjacent. According to this rule, we can model the topological relationship of the spatial distribution of voxel blocks as a graph structure. Inspired by the above, we introduce color histograms to define our rules from the perspective of the spectral characteristics of point clouds. The color histogram intersection is used to measure the similarity of the spectral distribution characteristics between two multispectral point cloud voxel blocks. Specifically, the color histogram intersection is computed by evaluating the overlap between the two histograms, $h_i$ and $h_j$, of the voxel blocks. The specific definition rules are as follows:

$$\mathbf{A}_{\text{ij}} = \begin{cases} 1, & \text{if } i \neq j \text{ and } s\left(h_i, h_j\right) > \theta \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$s\left(h_i, h_j\right) = \sum_k \min\left(h_i(k), h_j(k)\right) \tag{9}$$

where $s(h_i, h_j)$ represents the intersection of color histograms between pairs of voxels with $\theta = 0.5$, $h_i(k)$ and $h_j(k)$ represent the values in the $k$th color bin of histograms $h_i$ and $h_j$, respectively. We then use the Dijkstra algorithm to calculate the shortest path length between nodes in the two graph structures and set a maximum length constraint (set to the split number +2 in this article) to explicitly specify the path length category for subsequent network learning. Finally, we preprocess the shortest path length between voxel blocks in both 3-D Euclidean space and spectral space. Save to two adjacent matrices $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$ used as a follow-up self-supervision signal.

### C. Spatial–Spectral Distance Reconstruction Module

We propose the S-SDP module as the core module of S4DR-Net. The overall structure of the module is shown in Fig. 2. This module samples a dual-branch structure, guided by spatial and spectral information, respectively. First, we denote the extracted voxel features as $\mathrm{F} = [f_1, f_2, \ldots, f_V]$, $f_v \in \mathbb{R}^C$, where $C$ is the number of feature channels, and $V$ is the number of voxel blocks, and use it for graph modeling. The specific process is as follows:

$$E(u, v) = \text{Concat}\left(f_v - f_u, f_u\right), \quad E(u, v) \in \mathbb{R}^{2C\times V\times V} \tag{10}$$

where $E \in \mathbb{R}^{2C\times V\times V}$ represents the graph model. Under each branch, we employ multihead graph attention mechanisms to learn topological relationships between voxel blocks. The aggregated features are then mapped to path length categories using MLPs. During this process, the shortest path lengths generated in the data preprocessing stage $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$ serve as guidance labels, enabling the model to self-learn intrinsic topological relationships between pairwise voxel blocks in both Euclidean space and spectral space. Take the example of branch guidance based on spectral information in the module. The specific process is as follows:

$$\tilde{E}(u, v) = \mathrm{M} - \mathrm{hGAT}\left(E(u, v)\right), \quad \tilde{E}(u, v) \in \mathbb{R}^{2C\times V\times V} \tag{11}$$

$$\hat{\mathbf{Dis}}_{\text{spectral}}(u, v) = \text{Softmax}\Big(\text{MLP}\left(\tilde{\mathrm{E}}(\mathrm{u}, \mathrm{v})\right)\Big) \tag{12}$$

$$\text{Loss}_{\text{spectral}} = -\sum_{i=1}^{C} \mathbf{Dis}_{\text{spectral}} \log(\hat{\mathbf{Dis}}_{\text{spectral}}) \tag{13}$$

$$\text{Loss}_{\text{pretraining}} = \text{Loss}_{\text{spatial}} + \text{Loss}_{\text{spectral}} \tag{14}$$

where M-hGAT represents the multihead graph attention mechanism, which is explained in detail in Section III-D, and $\hat{\mathbf{Dis}}_{\text{spectral}}(u, v)$ represents the prediction based on spectral information. Similarly, based on spatial information, with $\mathbf{Dis}_{\text{spatial}}$, we can also get $\text{Loss}_{\text{spatial}}$. After each pass through the S-SDP module, the sum of dual-path prediction $\text{Loss}_{\text{pretraining}}$ is returned.

Notably, we again leverage $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$ to explicitly constrain feature extraction results. Since remote sensing scenarios involve numerous fine-grained classification tasks that heavily depend on local information, we expect the model to fully concentrate on learning local features. Meanwhile, according to the first law of geography, $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$ measure spatial correlations under different contexts: as the

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shortest path distance gradually increases, the model progressively shifts its focus to long-range global information. Therefore, based on the magnitudes of $\mathbf{Dis}_{\text{spatial}}$ and $\mathbf{Dis}_{\text{spectral}}$, we assign greater weights to short-distance edges. The specific procedure is given by the following formula:

$$\text{M}-\text{hGAT}\left(\mathbb{G}\left(\mathbf{Dis}_{\text{spectral}}\right)\odot\tilde{E}(u,v)\right) \tag{15}$$

where $\mathbb{G}(x)=(1/(2\pi)^{1/2})\exp(-(1/2\sigma^2)x^2)$ denotes a Gaussian function with $\alpha^2=1,\mu=0$. It should be noted that the version S-SDP_v1 we initially adopted employed the predicted distance matrix $\hat{\mathbf{Dis}}_{\text{spatial}}$ and $\hat{\mathbf{Dis}}_{\text{spectral}}$ as the constraint signal. Subsequent experiments, however, demonstrated that the solution currently S-SDP_v2 employed is more reasonable. A detailed discussion can be found in Section IV. The spatial information-guided branch undergoes parallel processing. Ultimately, the network outputs the sum of features extracted from both pathways.

### D. FE Module and Graph Attention Mechanism

In this work, we utilize an FE module to enhance the local neighborhood features in point clouds following KNN sampling. The specific process is outlined in the formula below

$$f_i^{k_1}=\text{KNN}(f_i) \tag{16}$$

$$\tilde{f}_i^{k_1}=\text{MLP}\left(\text{Concat}\left(f_i,f_i^{k_1}-f_i\right)\right) \tag{17}$$

$$\tilde{f}_i=\text{Maxpool}\left(\tilde{f}_i^{k_1}\right). \tag{18}$$

Simultaneously, we introduce a multihead graph attention mechanism into the S-SDP module to enhance the dynamic weight allocation capability of graph convolutional networks for neighboring node features. Through this multihead mechanism, the model can learn node relationships from multiple distinct subspaces, capturing richer and more fine-grained structural information. The detailed procedure is given by the following formulas:

$$\mathbf{H}^{(i)}\in\mathbb{R}^{d\times N\times N},\quad i=1,2,\ldots,h \tag{19}$$

$$\widetilde{\mathbf{H}}^{(i)}=\text{Softmax}(\text{Conv3}d(\phi(\mathbf{H}^{(i)})))\odot\mathbf{H}^{(i)} \tag{20}$$

$$\widetilde{\mathbf{H}}=\text{Concat}\left(\widetilde{\mathbf{H}}^{(1)},\widetilde{\mathbf{H}}^{(2)},\ldots,\widetilde{\mathbf{H}}^{(h)}\right)\in\mathbb{R}^{C\times N\times N} \tag{21}$$

where $\mathbf{H}\in\mathbb{R}^{C\times N\times N}$ represents the graph structure, and $C$ denotes the number of edge feature channels and $N$ is the number of nodes. We uniformly partition the features along the channel dimension into $h$ heads, with each head having dimension $d=(C/h)$. The attention scores $\text{Softmax}(\text{Conv3}d(\phi(\mathbf{H}^{(i)})))$ are computed for each head and applied to their corresponding feature heads. Finally, the partitioned feature heads are merged to form the final result.

### E. Loss Function

The loss function for model fine-tuning includes the training set label prediction loss, which we measure using cross-entropy loss, as well as the distance reconstruction loss of the S-SDP module. The formula is as follows:

$$\text{Loss}_{\text{label}}=-\sum_{i=1}^{C}y_i\log(\hat{y}_i) \tag{22}$$

$$\text{Loss}=\text{Loss}_{\text{label}}+\text{Loss}_{\text{pretraining}} \tag{23}$$

where $\hat{y}_i$ denotes the prediction result for land categories, $\text{Loss}_{\text{pretraining}}$ denotes the loss of the S-SDP module for distance reconstruction.

## IV. Experiments

The experiments are structured into five stages. For the first part, we pretrain our model using the proposed method on the target dataset's training set. After loading the pretrained model weights, we perform fine-tuning with a standard point cloud classification head. Comparisons with recent state-of-the-art fully supervised methods demonstrate the effectiveness of our pretraining approach. For the second part, we compare against classical point cloud self-supervised pretraining frameworks, validating that our method offers superior applicability for multispectral point cloud classification in remote sensing scenarios. Thirdly, we conducted ablation experiments on the S-SDP module in S4DR-Net. In addition, by progressively reducing the label size of training data during fine-tuning tasks, we verify that our self-supervised pretraining approach significantly reduces model dependency on labeled data while maintaining performance. Finally, we conducted a detailed discussion on the rationality of the v1 and v2 versions of S-SDP. Through experimental verification, we found that S-SDP_v2 is more reasonable than S-SDP_v1. In addition, we further explored the impact of different parameter freezing strategies during the fine-tuning phase on the pretraining results.

All experiments were conducted on a computing server equipped with 8×NVIDIA[1] RTX 4090 GPUs (each with 24 GB GDDR6X memory). The host machine featured an Intel[1] Xeon[1] Gold 5318Y CPU @ 2.10 GHz and ran the Ubuntu 20.04.6 LTS operating system. The deep learning framework is based on PyTorch 2.1.1 (compiled with CUDA 11.8), with hardware acceleration enabled by the NVIDIA 535.98 driver. We uniformly selected the following metrics: per-class intersection over union (per-class IoU), mean intersection over union (mIoU), overall accuracy (OA), and average $F1$-score (Avg. $F1$).

### A. Description of the Data

We employed two real-world airborne multispectral LiDAR datasets, as illustrated in Fig. 3. Harbor of Tobermory Dataset (HT) contains 7 181 982 spatial points collected via the Optech Titan triple-channel LiDAR system [40]. This system acquires data in three spectral bands (532, 1064, and 1550 nm). To address multispectral data misalignment caused by varying sensor incidence angles, our study used the 1550-nm channel as the primary source, fusing intensity information from other bands through nearest neighbor interpolation. The dataset is manually labeled into nine classes of land cover: barren, building, car, grass, powerline, road, ship, tree, and water. Via the same system, the University of Houston Dataset (UH) contains 4 436 470 spatial points with synchronously fused triple-band intensities. Annotations include eight categories: barren, commercial building, car, grass, powerline, road, residential building, and tree. It is noteworthy that class imbalance

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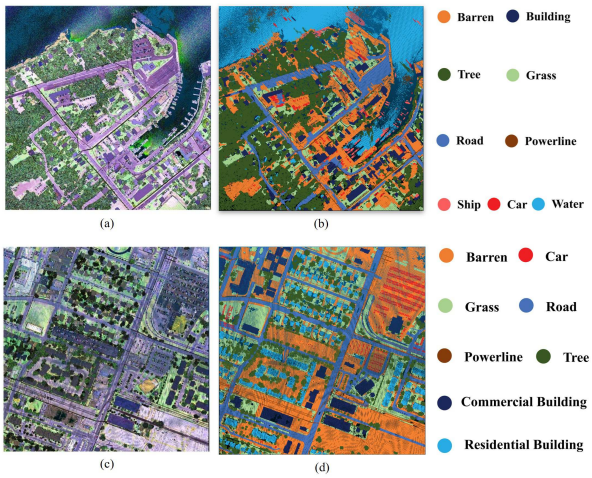



Fig. 3. Visualization of multispectral point clouds of two scenes. (a) Visualization of the harbor of Tobermory. (b) Label of the harbor of Tobermory. (c) Visualization of the University of Houston. (d) Label of the University of Houston.

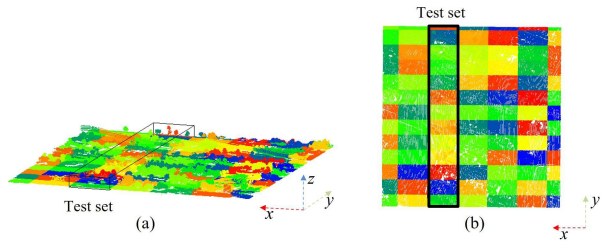



Fig. 4. Visualization of the specific division method of the University of Houston. (a) Visualization of the strabismus angle. (b) Visualization of bird-view. We use random color assignment for each divided point cloud block to visualize the division process of the point cloud.

represents a prevalent challenge for classification in remote sensing scenes. For these two specific datasets, the car and powerline classes exhibit particularly scarce samples compared to the other categories.

### B. Pretraining and Fine-Tuning

We pretrain the model using the training set within the same scene. Subsequently, we load the pretrained model and fine-tune the semantic classification head as illustrated in Fig. 1, where the blue region represents the pretrained model, and components outside this region constitute the semantic classification head.

To validate the effectiveness of our method, the dataset partitioning strategy is implemented as follows. Taking the UH dataset as an example shown in Fig. 4, we divide the dataset into seven equal segments along the $x$-axis direction of spatial points, forming seven distinct regions. For each area, data is further partitioned along the $y$-axis direction to facilitate efficient data loading. After evaluating the class distribution of ground objects across all areas, we select the area with the most balanced class distribution, Area 5, as our test set, while the remaining regions constitute the training set. It is noteworthy that an identical partitioning methodology is applied consistently to both datasets. To validate the performance of our method for multispectral point cloud classification in remote sensing scenarios, comparative experiments are conducted with state-of-the-art approaches across four paradigms: Mamba/transformer-based methods (PointTransformer [23], 3-D-Umamba [26], PointCloudMamba [25]); classical MLP-based architectures PointNet++ [20]; and graph-learning-based nontransductive method MKGSL [32].

*1) Classification Results on HT:* As detailed in Table I, our pretraining framework demonstrates exceptional performance across classification metrics. Our approach achieves an OA of 91.4%, with the mIoU reaching 84.6% and Avg. $F1$ attaining 91.4%. These results indicate robust land cover classification capabilities across diverse terrain categories. Fig. 5 is a confusion matrix diagram of the classification indicators for each method. It can be seen that our method not only has excellent classification results for large objects, but can also identify high-granularity objects. Fig. 6 shows the global visualization results for the test area, and Fig. 7 shows the local visualization results for Area A. The experimental observations are as follows.

1) As shown in Table I, the attention-based PointTransformer performs poorly on complex and irregular remote sensing point cloud datasets. This inspires us to conclude that for remote sensing scenarios with high-granularity classification requirements, we need sufficient local information rather than global information obtained through modeling based on global context dependencies.
2) From the visualization results, it can be seen that our pretraining method has excellent classification results for Car, indicating that our network has excellent understanding capabilities for high-granularity objects, meeting the classification requirements for high-granularity objects in remote sensing scenarios. At the same time, it also has excellent classification results for large buildings, demonstrating that our method has multiscale classification capabilities.

*2) Classification Results on UH:* Table II presents quantitative metrics evaluating classification performance across various land cover categories using different methods. As evidenced by the results, our self-supervised pretraining framework significantly enhances model performance for land cover classification in remote sensing scenarios. Particularly for sparsely distributed and fine-grained categories such as powerlines and cars, our network achieves substantially improved results. Fig. 8 displays confusion matrices for classification metrics across all methods, demonstrating that our network achieves optimal comprehensive classification performance. Fig. 9 provides a global visualization of the test area, while Figs. 10 and 11 offer localized visualizations for Area A and Area B, respectively. In addition to the observations on the HT dataset, we can also see that our model can better distinguish the boundaries between grass and barren, while ensuring the classification accuracy for highly granular ground features such as powerlines. Thus, our method can also better differentiate the boundaries of remote sensing ground objects.

### C. Comparison of Self-Supervised Point Cloud Pretraining Methods

In this section, we conducted comparative experiments between S4DR-Net and the classic point cloud self-supervised

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TABLE I

COMPARISON OF EVALUATION METRICS ON THE HT DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Building | Car | Grass | Powerline | Road | Ship | Tree | Water | | | |
| PointNet++ [20] | 65.5 | 79.6 | 89.5 | 94.2 | 87.1 | 67.2 | 65.3 | 74.4 | **97.0** | 88.5 | 88.4 | 80.0 |
| MKGSL [32] | 77.7 | 72.2 | 30.5 | 82.2 | 41.1 | 75.3 | 49.8 | **96.7** | 88.4 | 80.1 | 79.5 | 68.2 |
| PointTransformer [23] | 59.4 | 78.9 | 89.2 | 26.6 | 76.3 | 55.9 | 16.7 | 88.8 | 89.3 | 76.8 | 74.4 | 64.6 |
| PointCloudMamba [25] | **79.3** | **86.4** | **97.0** | 96.0 | **97.8** | **77.4** | 55.5 | 83.0 | 81.8 | 91.0 | 90.6 | 83.8 |
| 3D-UMamba [26] | 62.6 | 79.5 | 88.1 | 71.9 | 85.1 | 74.7 | 75.2 | 78.2 | **97.0** | 87.9 | 88.0 | 79.1 |
| S4DR-Net(Ours) | 73.5 | 85.7 | 86.2 | **98.0** | 87.0 | 74.5 | **80.3** | 80.9 | 95.1 | **91.4** | **91.4** | **84.6** |

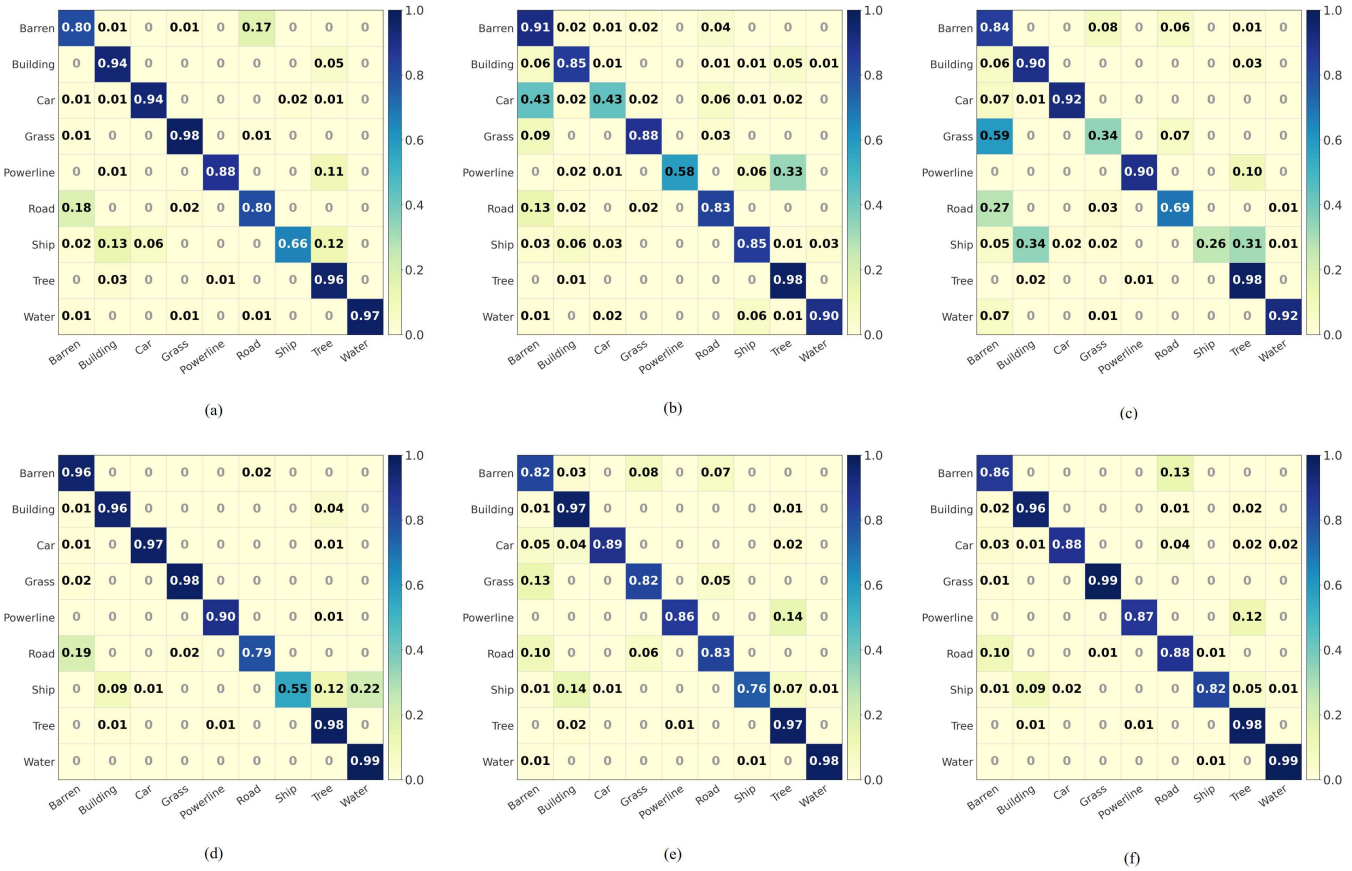



Fig. 5. Confusion matrix of classification results on the HT dataset. (a) PointNet++. (b) MKGSL. (c) PointTransformer. (d) PointCouldMamba. (e) 3-D-UMamba. (f) S4DR-Net (Ours).

pretraining framework to verify its effectiveness on multispectral point clouds in remote sensing scenarios. We conducted comparative analyses with methods, including Point2Vec [41], Point-MAE [34], IAE [42], and Point-BERT [35] on two datasets. To ensure fairness, we used one of the two datasets as the pretraining dataset for each framework. Specifically, for the HT dataset, we used the UH dataset as the pretraining base dataset, then fine-tuned the pretrained model on the training set of the HT dataset, and finally validated it on the test set. Similarly, for the UH dataset, we used the HT dataset for pretraining.

As shown in Tables III and IV, the experimental results reveal that our pretrained model consistently outperforms existing methods, particularly for high-granularity objects like cars and buildings. Notably, on the HT dataset, our pretrained model achieves the highest performance for categories such as building (79.9%) and car (85.4%) compared to Point2Vec, Point-MAE, and others. Similarly, on the UH dataset, commercial building (78.9%) and road (61.4%) also show superior performance with our framework. These results demonstrate that our pretrained model not only achieves better classification performance for large land cover classes but also performs effectively on high-granularity land cover types, handling multispectral data more efficiently and outperforming baseline methods in both per-class IoU and OA.

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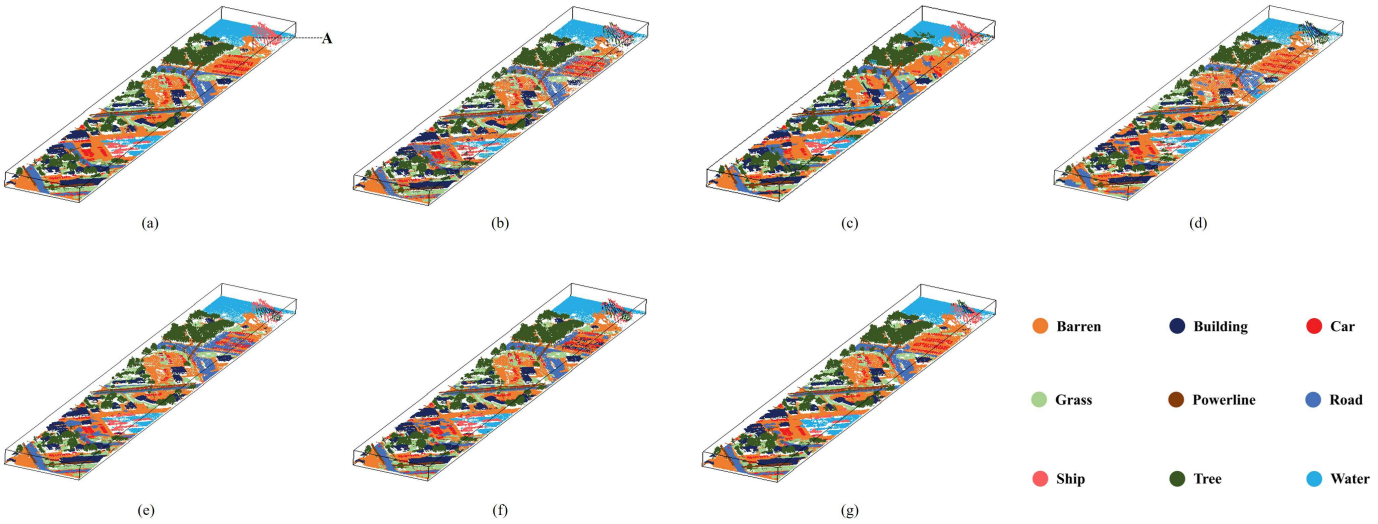



Fig. 6. Visualization of classification results on the HT dataset. (a) Ground truth. (b) PointNet++. (c) MKGSL. (d) PointTransformer. (e) PointCouldMamba. (f) 3-D-UMamba. (g) S4DR-Net (Ours).

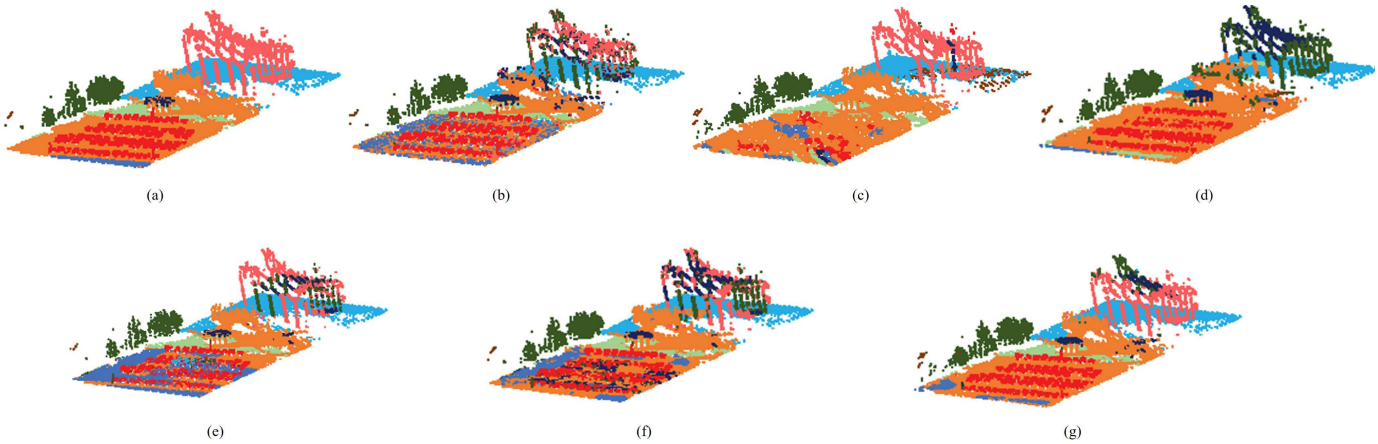



Fig. 7. Visualization of the classification results for Area A in the HT dataset. (a) Ground truth. (b) PointNet++. (c) MKGSL. (d) PointTransformer. (e) PointCouldMamba. (f) 3-D-UMamba. (g) S4DR-Net (Ours).

TABLE II

COMPARISON OF EVALUATION METRICS ON THE UH DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Car | Commercial | Grass | Road | Powerline | Residential | Tree | | | |
| PointNet++ [20] | 45.0 | **70.7** | 59.1 | 78.7 | 55.3 | 45.6 | 48.2 | 63.5 | 73.1 | 73.0 | 58.3 |
| MKGSL [32] | **65.3** | 16.1 | 65.5 | 67.9 | 55.4 | 20.5 | 68.8 | 86.5 | 79.7 | 69.1 | 55.8 |
| PointTransformer [23] | 43.4 | 52.4 | 74.0 | 83.8 | 60.8 | 71.7 | 63.2 | 80.3 | 78.4 | 78.9 | 66.2 |
| PointCloudMamba [25] | 44.9 | 23.0 | 77.0 | **84.6** | 64.0 | **83.0** | 75.8 | 71.2 | 78.0 | 77.0 | 65.4 |
| 3D-UMamba [26] | 42.7 | 57.4 | 70.0 | 67.5 | 56.0 | 76.7 | 65.2 | 83.7 | 77.4 | 78.0 | 64.9 |
| S4DR-Net(Ours) | 48.0 | 53.9 | **85.0** | 70.6 | **65.5** | 82.0 | **79.0** | **89.0** | **82.2** | **82.7** | **71.6** |

### *D. Ablation Experiments*

On the HT dataset, Table V presents our ablation experiments on the S-SDP module in the pretraining framework. In this ablation experiment, we removed two self-supervised signals $\mathcal{L}_{\text{spe}}$ and $\mathcal{L}_{\text{spa}}$ from the S-SDP module and then removed both self-supervised signals at the same time, using only the label loss to guide model training. First, as can be seen from Table V, our self-supervised pretraining model can guide the model to learn the implicit topological relationships inherent in multispectral point cloud data, thereby improving the model's generalizability. At the same time, it is worth noting that, compared to spatial information, the improvement in

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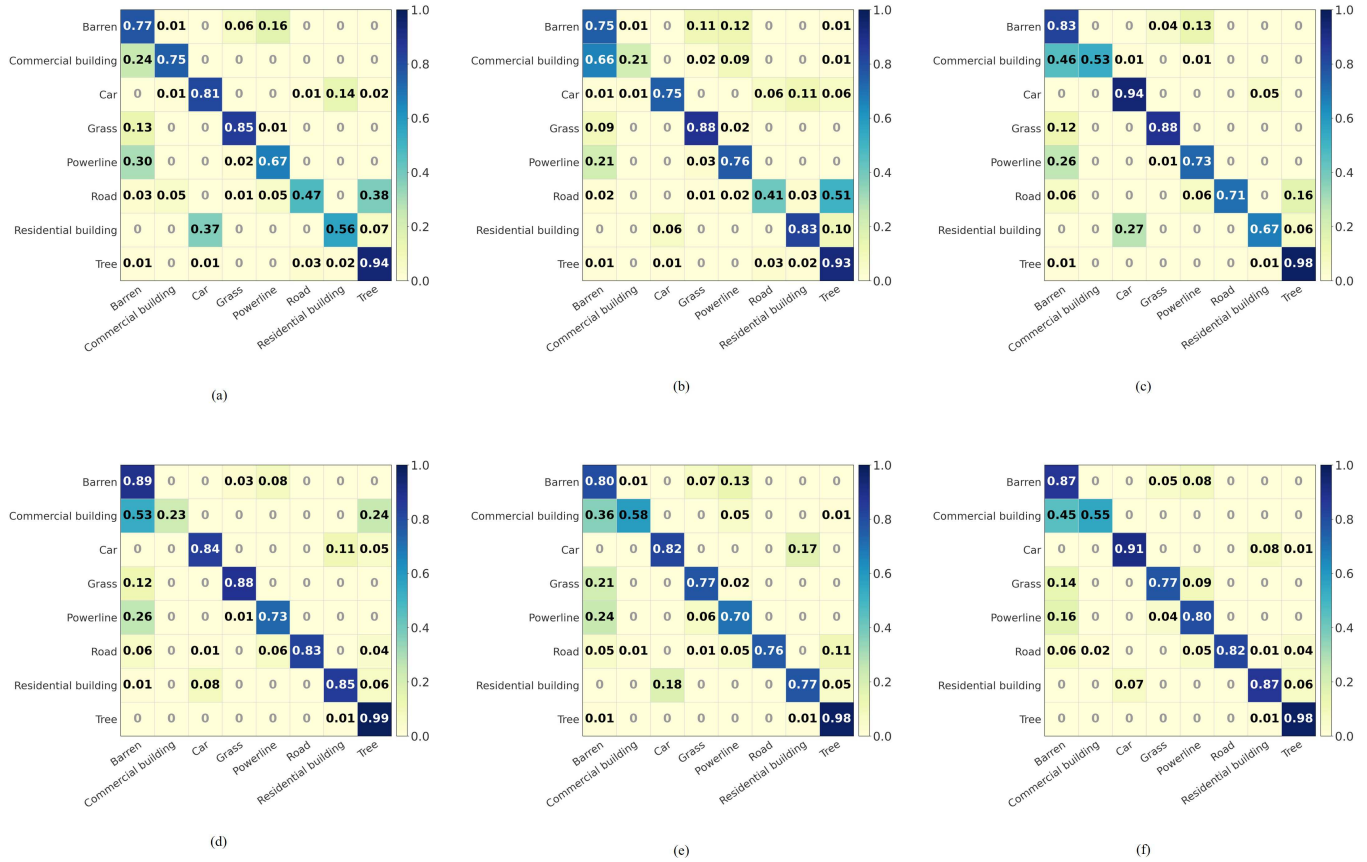



Fig. 8. Confusion matrix of classification results on the UH dataset. (a) PointNet++. (b) MKGSL. (c) PointTransformer. (d) PointCouldMamba. (e) 3-D-UMamba. (f) S4DR-Net (Ours).

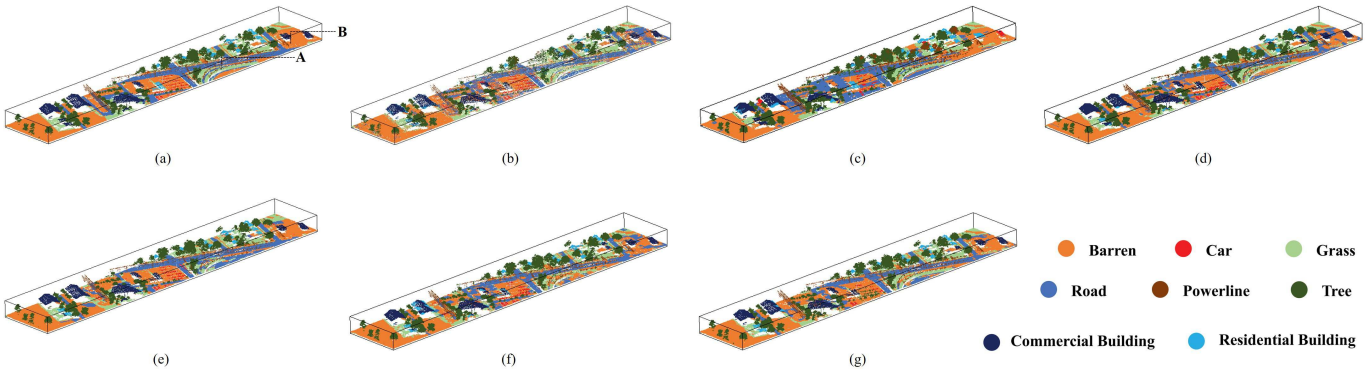



Fig. 9. Visualization of classification results on the UH dataset. (a) Ground truth. (b) PointNet++. (c) MKGSL. (d) PointTransformer. (e) PointCouldMamba. (f) 3-D-UMamba. (g) S4DR-Net (Ours).

spectral information supervision is more significant. Similarly, Table VI details ablation studies conducted on the UH dataset, revealing that in multispectral point cloud classification scenarios, self-supervised signals derived from spectral information contribute more significantly to performance gains than spatial information.

### *E. Label-Efficient Training With Limited Annotation*

To validate that our pretrained model reduces dependence on labeled data during fine-tuning—achieving the core goal of pretraining—we conduct experiments with varying fine-tuning label scales: 10%, 20%, 40%, and 80%. The results in Table VII demonstrate that our model retains robust performance even as the label scale decreases.

With 100% labeled data, the model achieves high mIoU scores of 83.6% on HT and 68.3% on UH. As the label scale drops to 80%, performance slightly decreases to 68.3% for HT and 51.0% for UH, indicating strong generalization despite less labeled data. At 40% labeled data, mIoU decreases to 54.0% on HT and 42.9% on UH, but the model still maintains reasonable accuracy for several classes, such as building and grass. With 20% labeled data, performance

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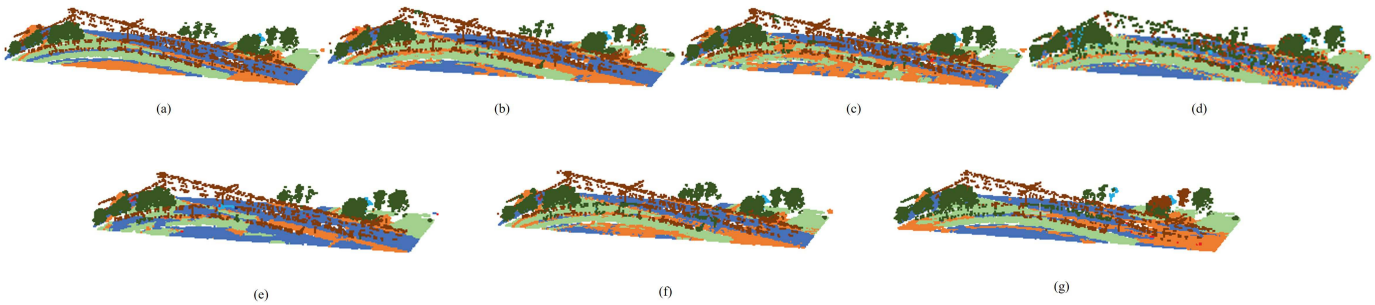



Fig. 10. Visualization of the classification results for Area A in the UH dataset. (a) Ground truth. (b) PointNet++. (c) MKGSL. (d) PointTransformer. (e) PointCouldMamba. (f) 3-D-UMamba. (g) S4DR-Net (Ours).

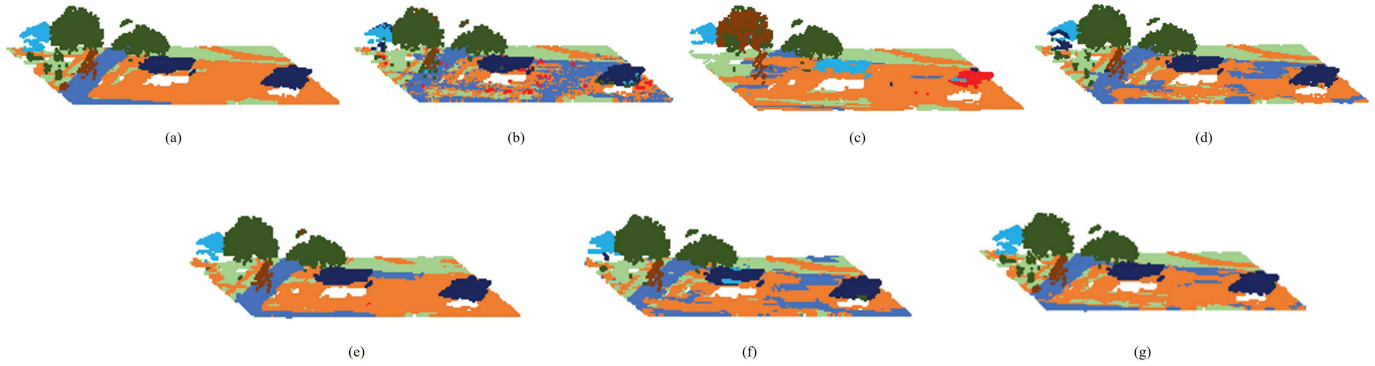



Fig. 11. Visualization of the classification results for Area B in the UH dataset. (a) Ground truth. (b) PointNet++. (c) MKGSL. (d) PointTransformer. (e) PointCouldMamba. (f) 3-D-UMamba. (g) S4DR-Net (Ours).

TABLE III

COMPARISON OF SELF-SUPERVISED POINT CLOUD PRETRAINING METHODS ON THE HT DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Building | Car | Grass | Powerline | Road | Ship | Tree | Water | | | |
| Point2Vec [41] | 49.3 | 65.2 | 9.3 | **35.1** | 62.0 | **48.1** | 32.2 | **92.3** | **96.5** | 64.0 | 66.5 | 54.5 |
| Point-MAE [34] | 45.4 | 53.9 | 49.1 | 8.3 | 56.8 | 15.9 | 33.0 | 67.3 | 59.8 | 58.6 | 57.6 | 43.3 |
| IAE [42] | 48.1 | 56.2 | 52.3 | 9.7 | 60.1 | 18.2 | **35.4** | 70.5 | 62.1 | 61.5 | 60.2 | 46.7 |
| Point-BERT [35] | 51.7 | 46.4 | 50.9 | 33.2 | 57.1 | 15.6 | 28.9 | 79.8 | 86.7 | 72.1 | 69.7 | 50.0 |
| S4DR-Net(Ours) | **56.0** | **79.9** | **85.4** | 26.9 | **72.7** | 47.5 | 23.6 | 89.2 | 91.3 | **75.1** | **74.2** | **63.3** |

TABLE IV

COMPARISON OF SELF-SUPERVISED POINT CLOUD PRETRAINING METHODS ON THE UH DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Car | Commercial | Grass | Road | Powerline | Residential | Tree | | | |
| Point2Vec [41] | 66.5 | 32.6 | 42.1 | 20.2 | 33.0 | 63.4 | 43.4 | 85.7 | 69.8 | 62.7 | 48.4 |
| Point-MAE [34] | 50.3 | 56.1 | 30.0 | 29.8 | 21.7 | 57.0 | 16.1 | 79.1 | 60.6 | 57.0 | 42.6 |
| IAE [42] | 55.8 | **61.3** | 41.2 | 39.6 | 28.1 | 61.7 | 21.9 | 82.6 | 64.1 | 61.9 | 47.8 |
| Point-BERT [35] | 68.0 | 22.2 | 69.2 | 33.6 | 45.2 | **65.4** | 52.6 | **91.7** | 75.8 | 69.4 | 56.0 |
| S4DR-Net(Ours) | **72.0** | 52.7 | **78.9** | **51.3** | **61.4** | 43.9 | **56.7** | 90.0 | **80.6** | **80.0** | **63.4** |

further declines, though classes like building (81.1%) on HT and tree (73.2%) on UH remain stable. With only 10% labeled data, the mIoU drops to 30.3% for HT, but the model still performs reasonably well on certain categories, such as tree. These results highlight that the pretrained model remains effective with limited labeled data, show-

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TABLE V

EVALUATION METRICS (%) FOR ABLATION EXPERIMENTS ON THE HT DATASET. THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Group | Settings | | Per-class IoU | | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\mathcal{L}_{spe}$ | $\mathcal{L}_{spa}$ | Barren | Building | Car | Grass | Powerline | Road | Ship | Tree | Water | | | |
| I | × | × | 58.1 | 82.0 | **86.6** | 27.1 | **73.6** | 52.5 | 23.1 | 90.9 | 82.1 | 76.1 | 73.4 | 64.0 |
| II | √ | × | **83.6** | 86.0 | 79.9 | **97.1** | 73.1 | **68.6** | 75.4 | 93.8 | 95.4 | 91.3 | 90.7 | 83.7 |
| III | × | √ | 78.2 | **88.5** | 82.0 | 97.0 | 72.0 | 61.9 | **79.8** | **94.6** | 96.9 | **92.2** | 90.5 | 83.4 |
| IV | √ | √ | 81.8 | 87.7 | 84.8 | 97.0 | 73.1 | 64.5 | 74.4 | 94.5 | **97.0** | 91.8 | **90.8** | **84.0** |

TABLE VI

EVALUATION METRICS (%) FOR ABLATION EXPERIMENTS ON THE UH DATASET. THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Group | Settings | | Per-class IoU | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $\mathcal{L}_{spe}$ | $\mathcal{L}_{spa}$ | Barren | Car | Commercial | Grass | Road | Powerline | Residential | Ship | | | |
| I | × | × | 68.8 | **43.8** | 78.6 | 80.0 | 58.6 | 68.1 | 40.5 | 94.2 | 76.2 | 78.6 | 66.6 |
| II | √ | × | 65.7 | 45.3 | 84.5 | 80.1 | 58.3 | 69.8 | 49.2 | 94.7 | 77.9 | 80.1 | 68.4 |
| III | × | √ | **69.4** | 39.1 | 70.0 | **82.0** | **58.9** | **73.2** | 42.5 | **95.3** | 76.9 | 79.0 | 67.4 |
| IV | √ | √ | 68.2 | 43.6 | **86.7** | 80.7 | 58.1 | 69.0 | **54.5** | 94.8 | **83.2** | **80.9** | **69.5** |

TABLE VII

PERFORMANCE COMPARISON OF HT AND UH DATASETS WITH VARYING TRAINING LABEL SCALES (%)

| Settings | HT | | | | | | | | | | UH | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | mIoU | barren | building | car | grass | power line | road | ship | tree | water | mIoU | barren | car | commercial | grass | road | power line | residential | tree |
| 100% | 83.6 | 80.4 | 90.2 | 82.4 | 97.5 | 72.7 | 62.1 | 76.0 | 94.2 | 96.6 | 68.3 | 66.4 | 42.0 | 84.1 | 81.4 | 54.6 | 71.8 | 50.8 | 95.0 |
| 80% | 68.3 | 68.3 | 70.7 | 24.8 | 79.2 | 43.2 | 65.8 | 37.4 | 95.6 | 95.5 | 51.0 | 43.5 | 17.8 | 60.7 | 69.9 | 43.1 | 27.0 | 62.4 | 83.3 |
| 40% | 54.0 | 49.9 | 76.4 | 46.6 | 29.7 | 66.5 | 41.8 | 23.4 | 89.4 | 62.1 | 42.9 | 47.7 | 12.1 | 41.0 | 63.8 | 44.7 | 15.4 | 44.1 | 74.4 |
| 20% | 42.3 | 32.9 | 81.1 | 59.7 | 25.9 | 19.1 | 31.3 | 1.2 | 74.2 | 55.4 | 38.8 | 35.4 | 11.5 | 26.1 | 63.0 | 41.5 | 18.2 | 41.6 | 73.2 |
| 10% | 30.3 | 29.7 | 45.9 | 36.8 | 21.3 | 11.4 | 29.5 | 0.2 | 56.4 | 47.1 | 36.5 | 60.8 | 2.2 | 20.7 | 7.7 | 29.5 | 51.2 | 25.4 | 94.8 |

TABLE VIII

PERFORMANCE COMPARISON OF S4DR-NET WITH DIFFERENT S-SDP VERSIONS AND PRETRAINED MODEL PARAMETER FREEZING STRATEGIES ON THE HT DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Building | Car | Grass | Powerline | Road | Ship | Tree | Water | | | |
| S4DR-Net_v1/v1 | 79.9 | 87.5 | 75.3 | 97.3 | 73.8 | 62.5 | 78.9 | 92.7 | **97.6** | 91.2 | 90.1 | 82.9 |
| S4DR-Net_v2/v1 | 80.4 | **90.2** | 82.4 | 97.5 | 72.7 | 62.1 | 76.0 | **94.2** | 96.6 | **91.4** | 90.6 | 83.6 |
| S4DR-Net_v1/v2 | **80.7** | 86.0 | 81.5 | 97.4 | 69.6 | 62.0 | 69.1 | 94.0 | 97.3 | 91.1 | 89.5 | 81.9 |
| S4DR-Net_v2/v2 | 73.5 | 85.7 | **86.2** | **98.0** | **87.0** | **74.5** | **80.3** | 80.9 | 95.1 | **91.4** | **91.4** | **84.6** |

casing its potential for label-efficient training in real-world applications.

### F. Discussion

In this discussion, we discuss two pivotal issues as follows.

1) Which structural design is more reasonable between the two distinct versions of the proposed S-SDP module.
2) What parameter freezing strategy should be adopted after obtaining the pretrained weighted model to fully leverage the generalization capability of the pretrained model?

In Tables VIII and IX, the naming method S4DR-Net_vx/vy encodes two technical dimensions: the first vx denotes the version of the S-SDP module (v1: baseline and v2: enhanced), while the second vy indicates pretrained model parameters freezing (v1: frozen and v2: unfrozen). As shown in Table VIII, the global metrics of v2/v2 achieves 84.6% mIoU (+1.0% versus v2/v1), with OA and Avg. $F1$ both peaking at 91.4%. Powerline IoU surges to 87.0% (+17.4% max

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TABLE IX

PERFORMANCE COMPARISON OF S4DR-NET WITH DIFFERENT S-SDP VERSIONS AND PRETRAINED MODEL PARAMETER FREEZING STRATEGIES ON THE UH DATASET (%). THE BEST METRICS ARE SHOWN IN BOLD, WHILE THE SECOND-BEST METRICS ARE INDICATED WITH AN UNDERLINE

| Method | Per-class IoU | | | | | | | | OA | Avg. F1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Barren | Car | Commercial | Grass | Road | Powerline | Residential | Tree | | | |
| S4DR-Net_v1/v1 | 66.4 | 43.8 | 80.6 | **81.9** | 55.7 | 69.6 | 43.6 | **95.0** | 82.0 | 78.9 | 67.1 |
| S4DR-Net_v2/v1 | 66.4 | 42.0 | 84.1 | 81.4 | 54.6 | 71.8 | 50.8 | **95.0** | 82.1 | 79.9 | 68.3 |
| S4DR-Net_v1/v2 | **69.1** | 38.1 | 82.7 | 81.4 | 57.5 | 68.4 | 46.1 | **95.0** | **83.2** | 78.9 | 67.3 |
| S4DR-Net_v2/v2 | 48.0 | **53.9** | **85.0** | 70.6 | **65.5** | **82.0** | **79.0** | 89.0 | 82.2 | **82.7** | **71.6** |

gain), while car accuracy reaches 86.2% (+3.8% versus v1/v1), validating enhanced small-object discrimination. As shown in Table IX, 71.6% mIoU for v2/v2 (+3.3% versus alternatives), with residential IoU doubling (79.0% versus 43.6% in v1/v1) and powerline soaring to 82.0% (+12.4%). The refined feature decoupling in S4DR-Net_v2/v2, combined with adaptive fine-tuning of pretrained weights, substantially boosts classification accuracy for challenging targets (e.g., powerlines and urban structures), establishing this configuration as optimal for self-supervised point cloud learning.

## V. CONCLUSION

In this article, we propose the first self-supervised pretraining method, S4DR-Net, for multispectral point cloud classification in remote sensing scenarios. Through the S-SDP module, we construct separate graphs to capture the topological relationships among voxel blocks of multispectral point clouds in 3-D Euclidean and spectral spaces, while simultaneously reconstructing their distance relationships across these domains. This method drives the model to learn implicit, universal features of multispectral point clouds in a self-supervised manner. Comprehensive experiments and visualizations demonstrate that the S-SDP module serves as both a synergistic supervisory signal for fully supervised methods and enables the extraction of more detailed spatial features, leading to more accurate boundary classification. Simultaneously, compared to traditional self-supervised pretraining methods, S4DR-Net is less affected by class imbalance issues, exhibits superior generalization capability and robustness, and significantly reduces the pressure during the fine-tuning stage, mitigating the reliance on labels during fine-tuning.

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